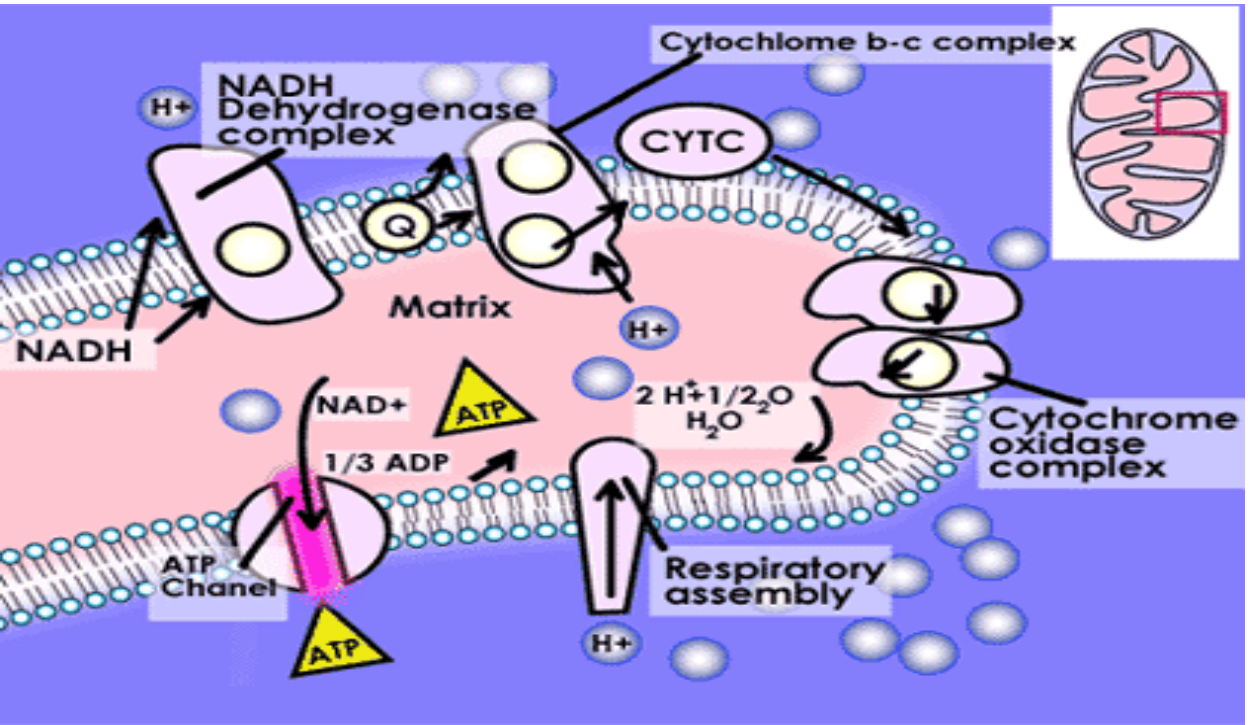
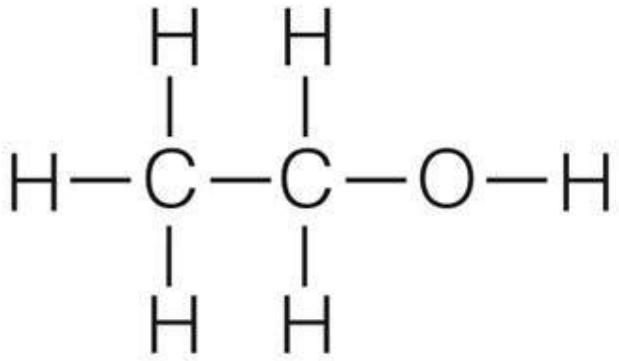
INTRODUCTION OF CARBOHYDRATE DIGESTION, ABSORPTION AND METABOLISM



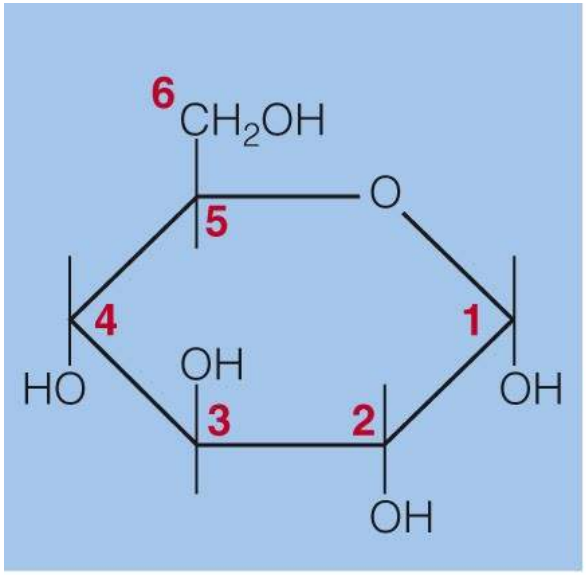
Carbohydrates

- Carbohydrates are made of carbon, hydrogen and oxygen atoms.
- These atoms form chemical bonds that follow the laws of nature.

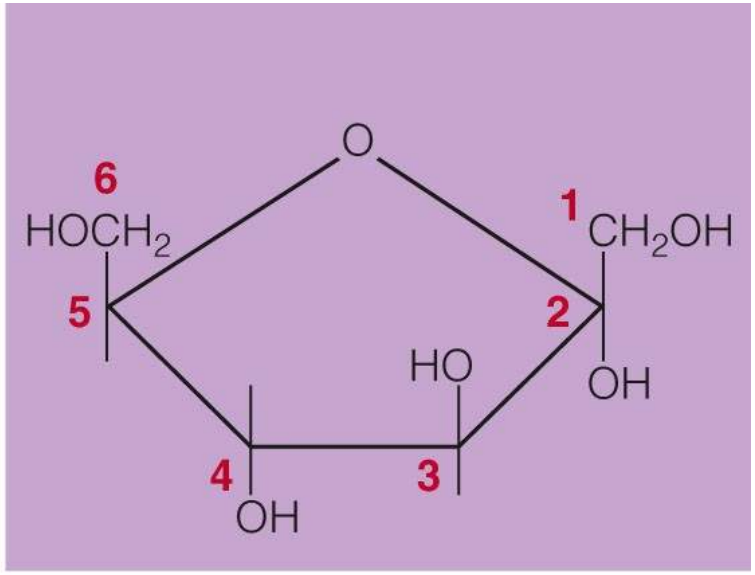


The Simple Carbohydrates

- Monosaccharrides are single sugars (most are hexoses).
  - Glucose serves as the essential energy source, and is commonly known as blood sugar or dextrose.
  - Fructose is the sweetest, occurs naturally in honey and fruits, and is added to many foods in the form of high-fructose corn syrup.
  - Galactose rarely occurs naturally as a single sugar.

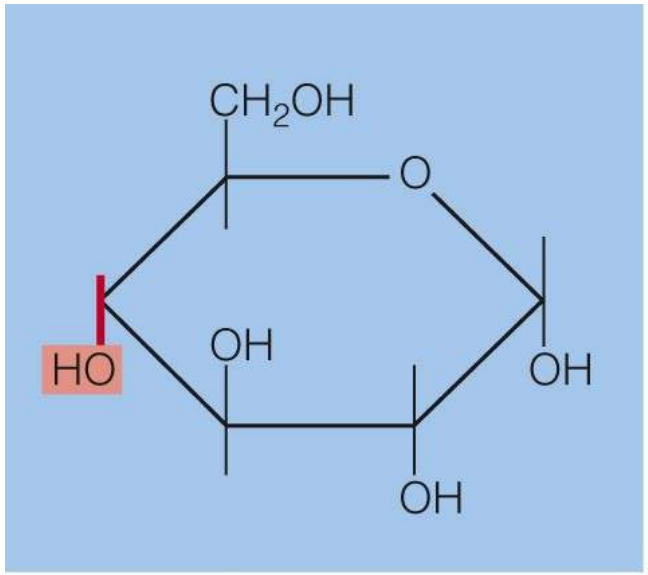


Glucose

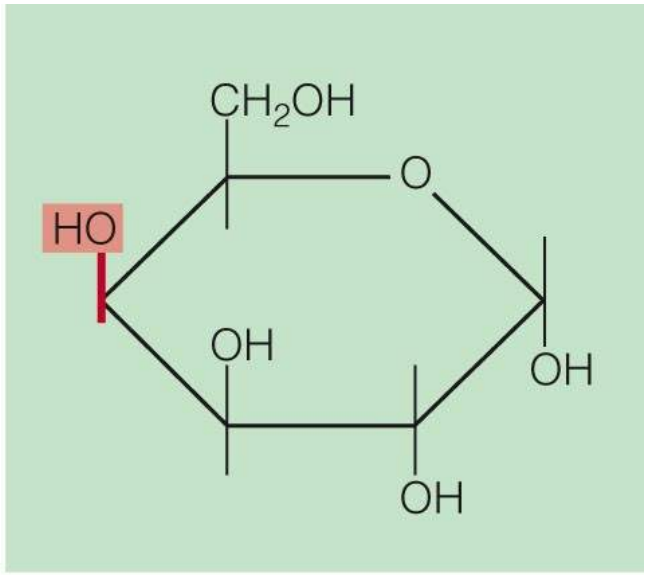


Fructose

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Glucose



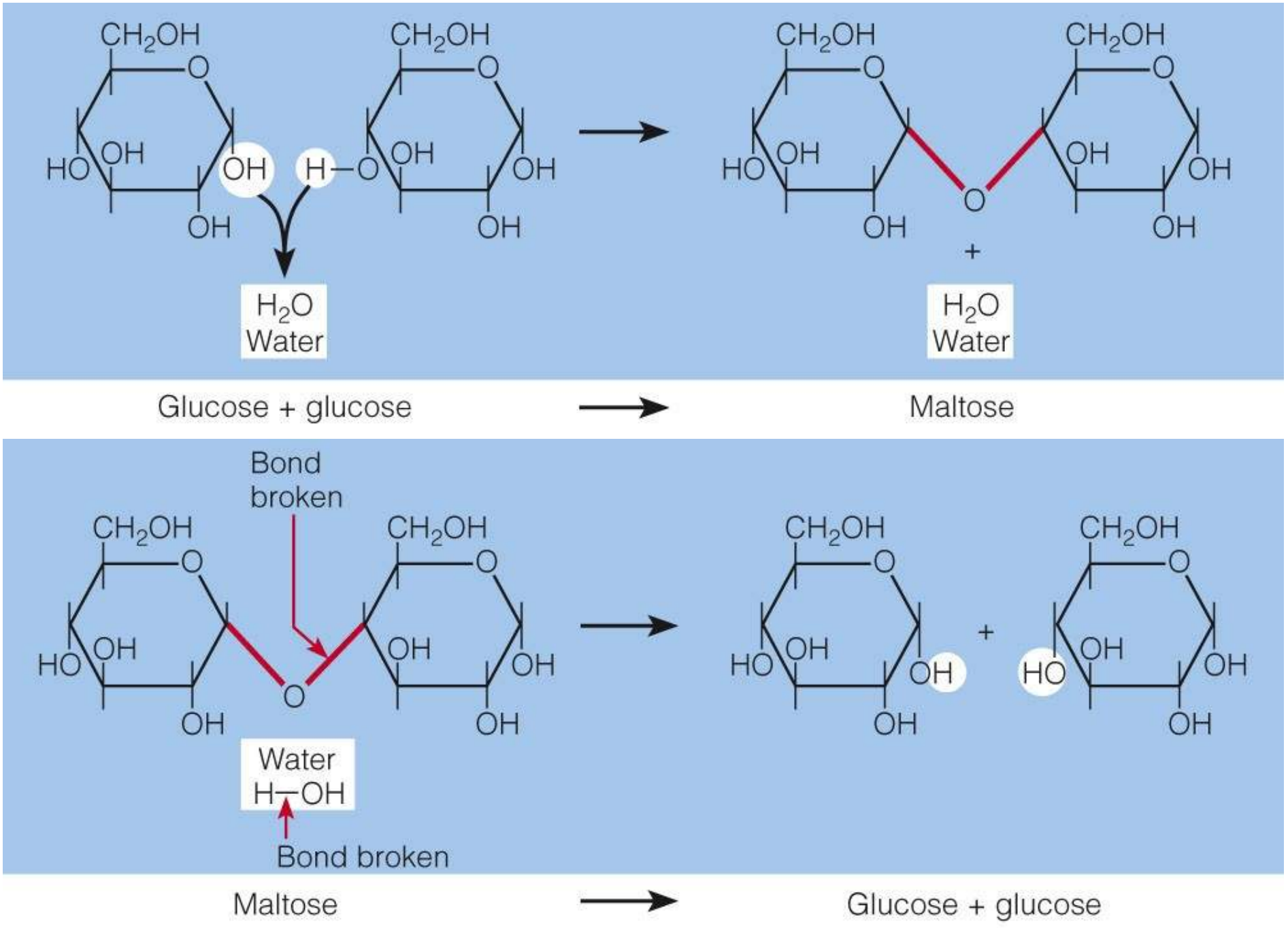
Galactose

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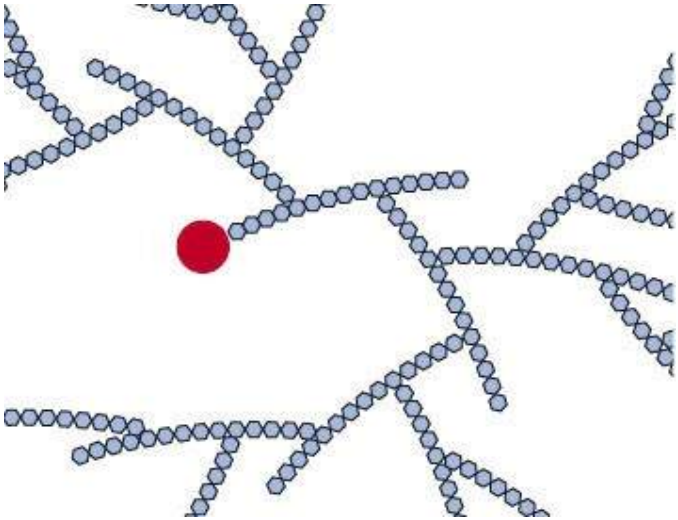
The Simple Carbohydrates

- Disaccharides are pairs of monosaccharides, one of which is always glucose
  - Condensation reactions link monosaccharides together.
  - Hydrolysis reactions split molecules and commonly occur during digestion.
  - Maltose consists of two glucose units. It is produced during the germination of seeds and fermentation.
  - Sucrose is fructose and glucose combined. It is refined from sugarcane and sugar beets and is readily available.
  - Lactose is galactose and glucose combined. It is found in milk and milk products.


CH_2OH
O
HO
OH
OH
OH
H—O
OH
OH
OH
H_2O
Water
+
H_2O
Water
Glucose + glucose
Maltose
Bond
broken
Water
H—OH
Bond broken
Maltose
Glucose + glucose

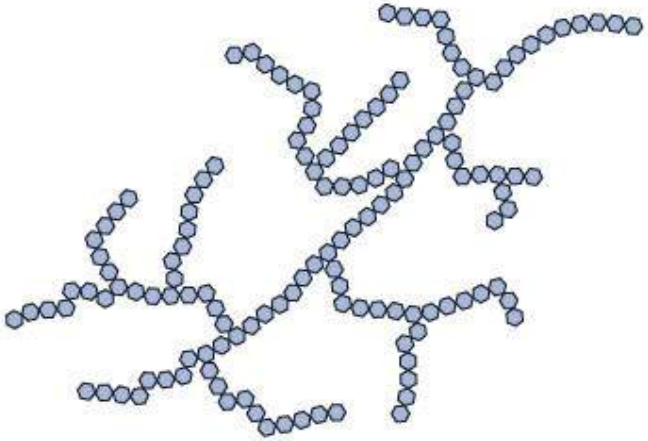
The Complex Carbohydrates

- Few (oligosaccharides) or many (polysaccharides) glucose units bound/linked together in straight or branched chains.
  Glycogen
  - Storage form of glucose in the body
  - Provides a rapid release of energy when needed
- **Starches**
  - Storage form of glucose in plants
  - Found in grains, tubers, and legumes

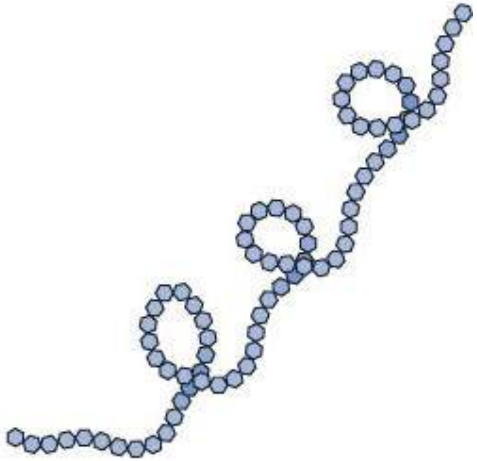


Glycogen

A glycogen molecule contains hundreds of glucose units in highly branched chains. Each new glycogen molecule needs a special protein for the attachment of the first glucose (shown here in red).



Starch (amylopectin)



Starch (amylose)

A starch molecule contains hundreds of glucose molecules in either occasionally branched chains (amylopectin) or unbranched chains (amylose).

Dietary Fiber



- Undigested plant food
- Beta glycosidic bond
- **Insoluble fiber**
  - Cellulose, hemicellulose, lignin
  - Not fermented by the bacteria in the colon
- **Soluble fiber**
  - Gum, Pectin, Mucilage
  - Fruit, vegetable, rice bran, psyllium seed

Forms of Carbohydrate

- Simple sugars
- Starch
- Glycogen
- Fiber


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Digestion of Carbohydrates

- Polysaccharides
- Disaccharides
- Monosaccharide
- (Absorption)

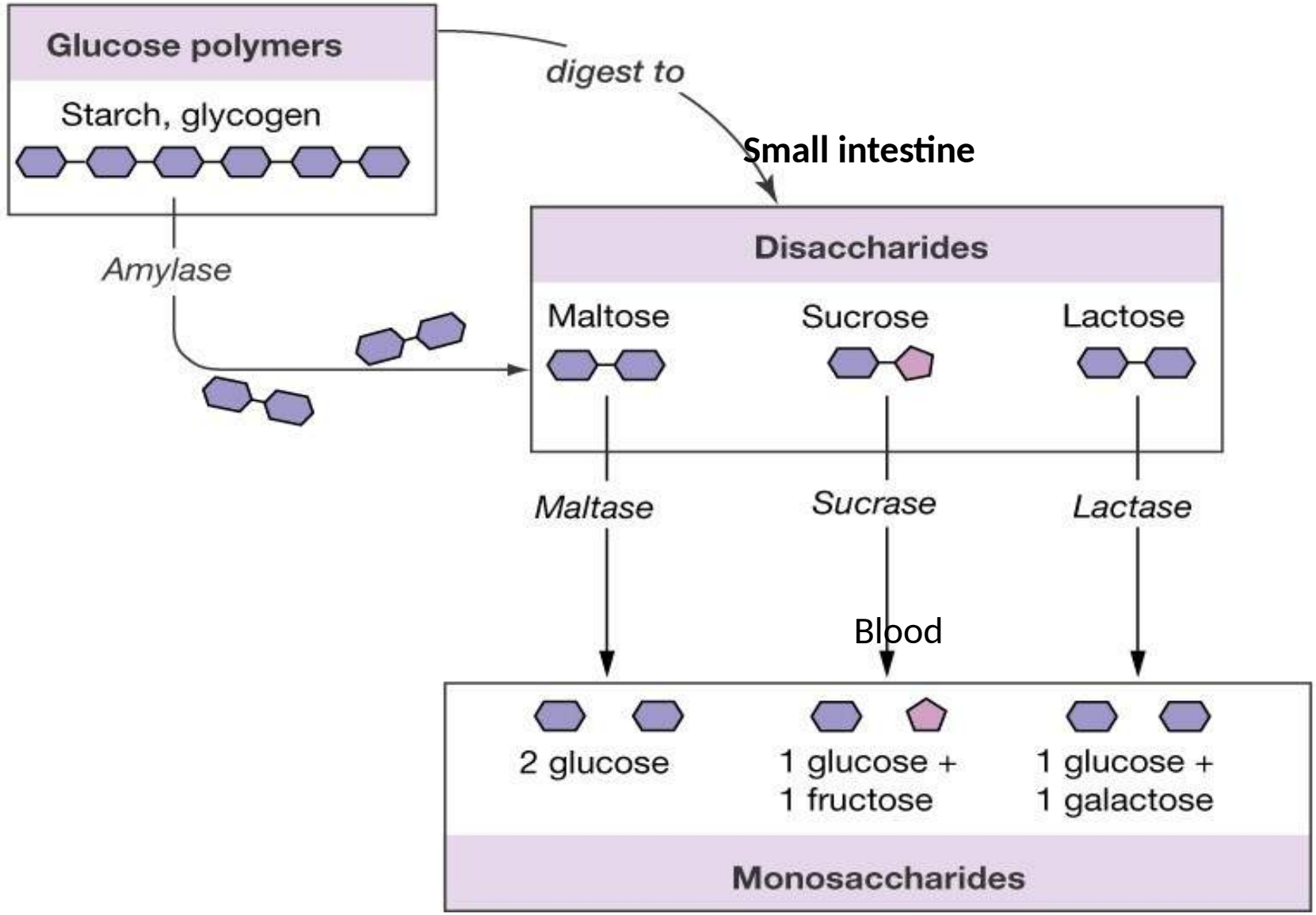
Starts in mouth

Completed in small intestine

animation

Nothing happens in the stomach because it is too acidic.

Digestion of Carbohydrates

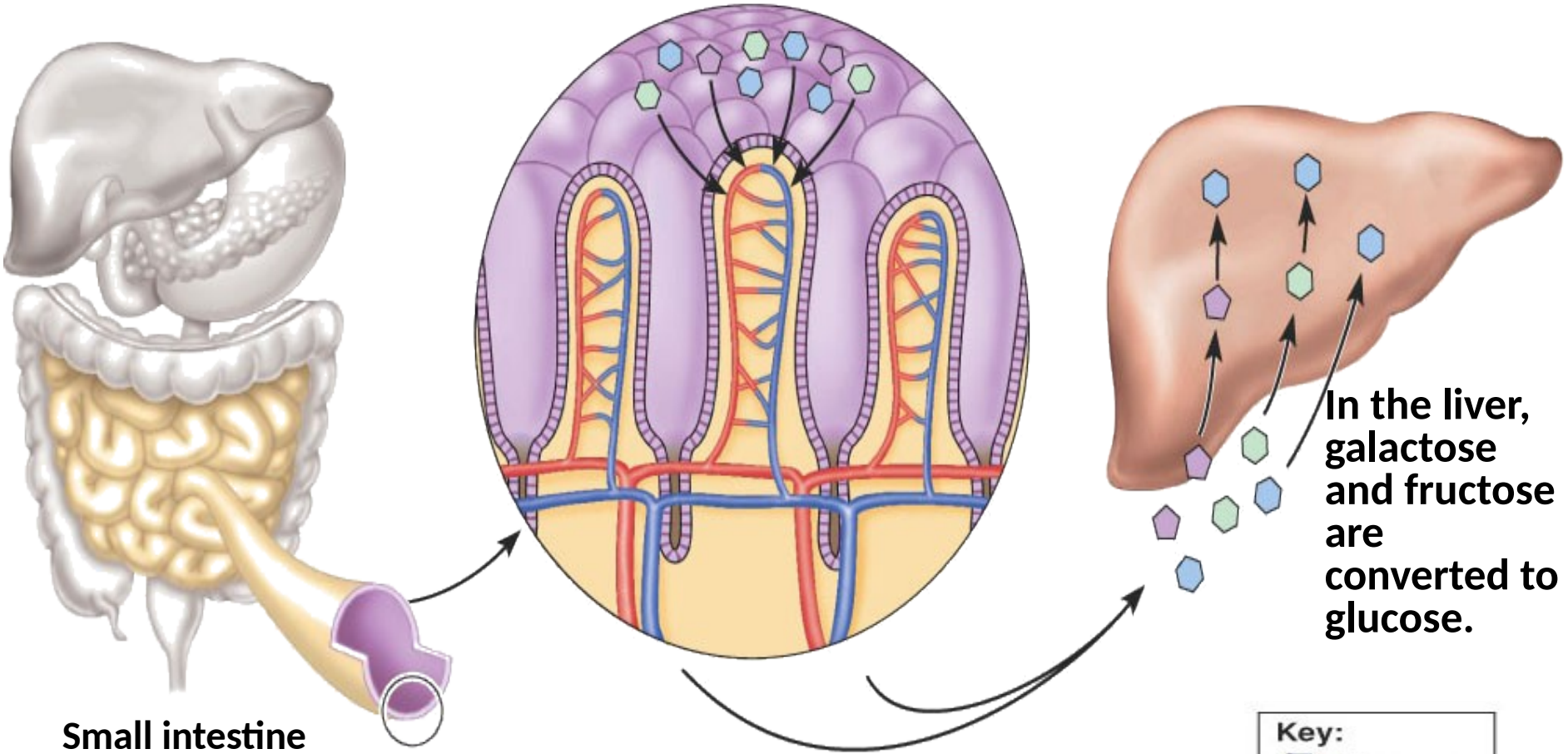


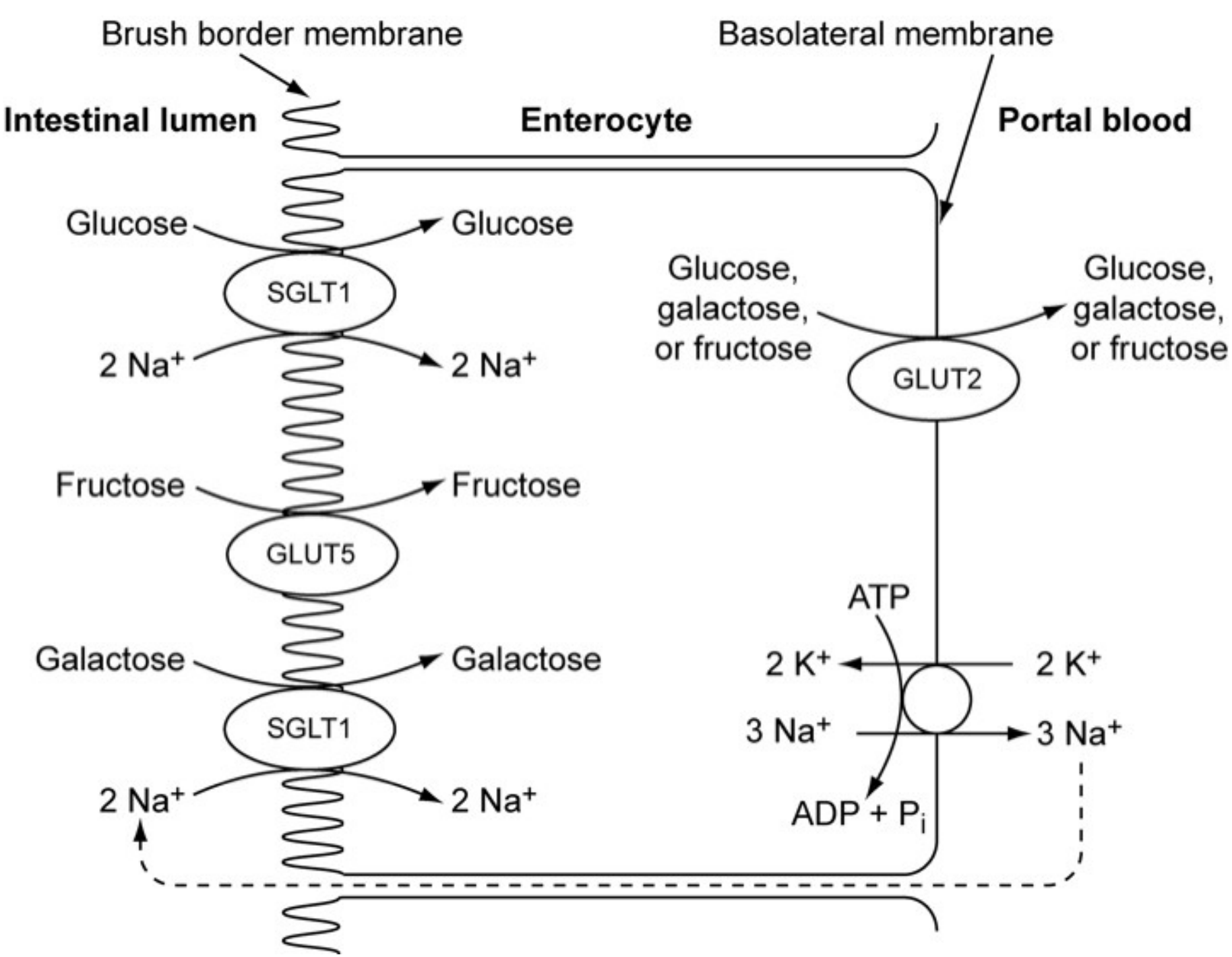
Digestion and Absorption of Carbohydrates

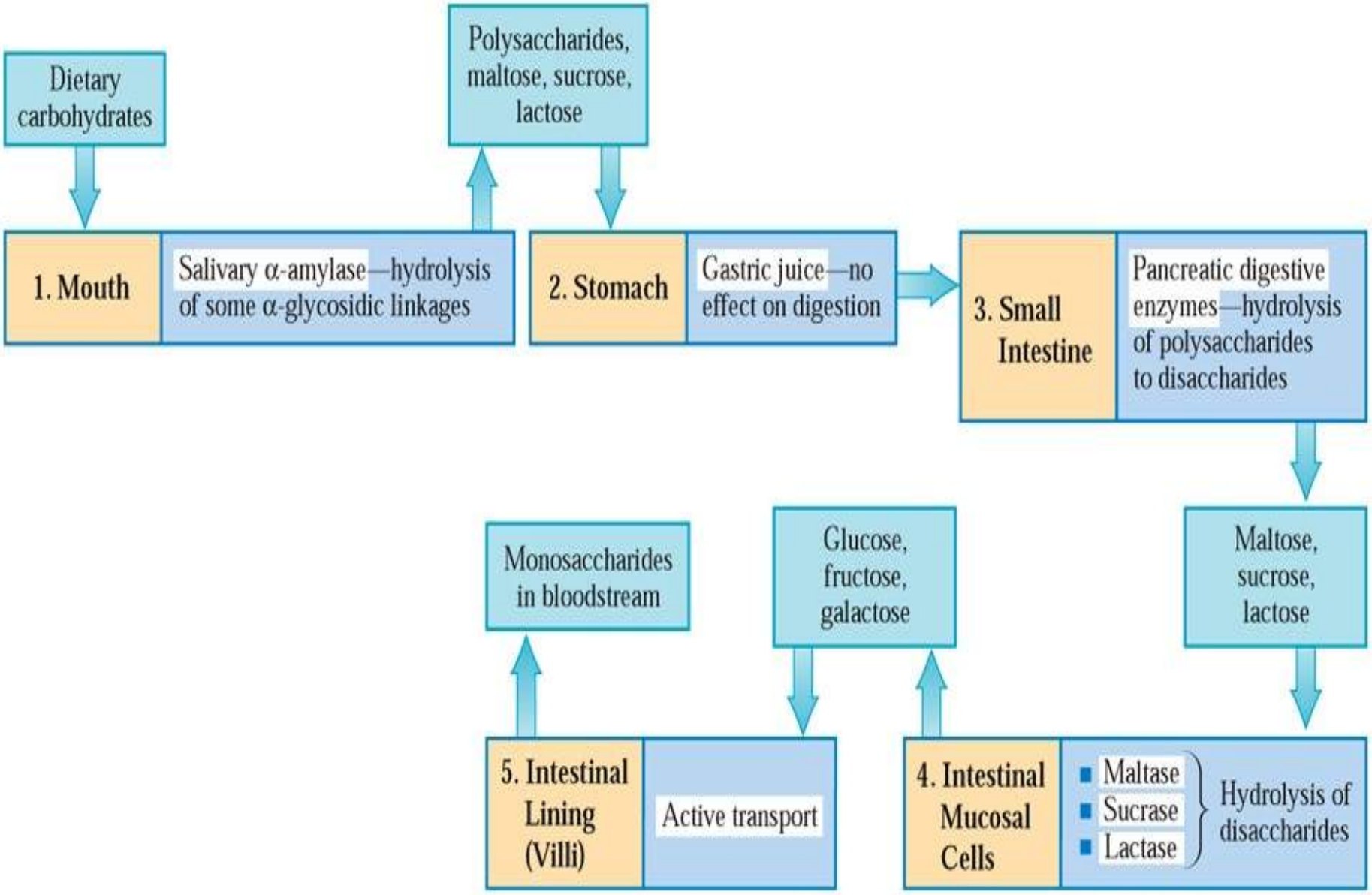
- Carbohydrate Absorption
  - Primarily takes place in the small intestine
  - Glucose and galactose are absorbed by **active transport.**
  - Fructose is absorbed by facilitated **diffusion.**

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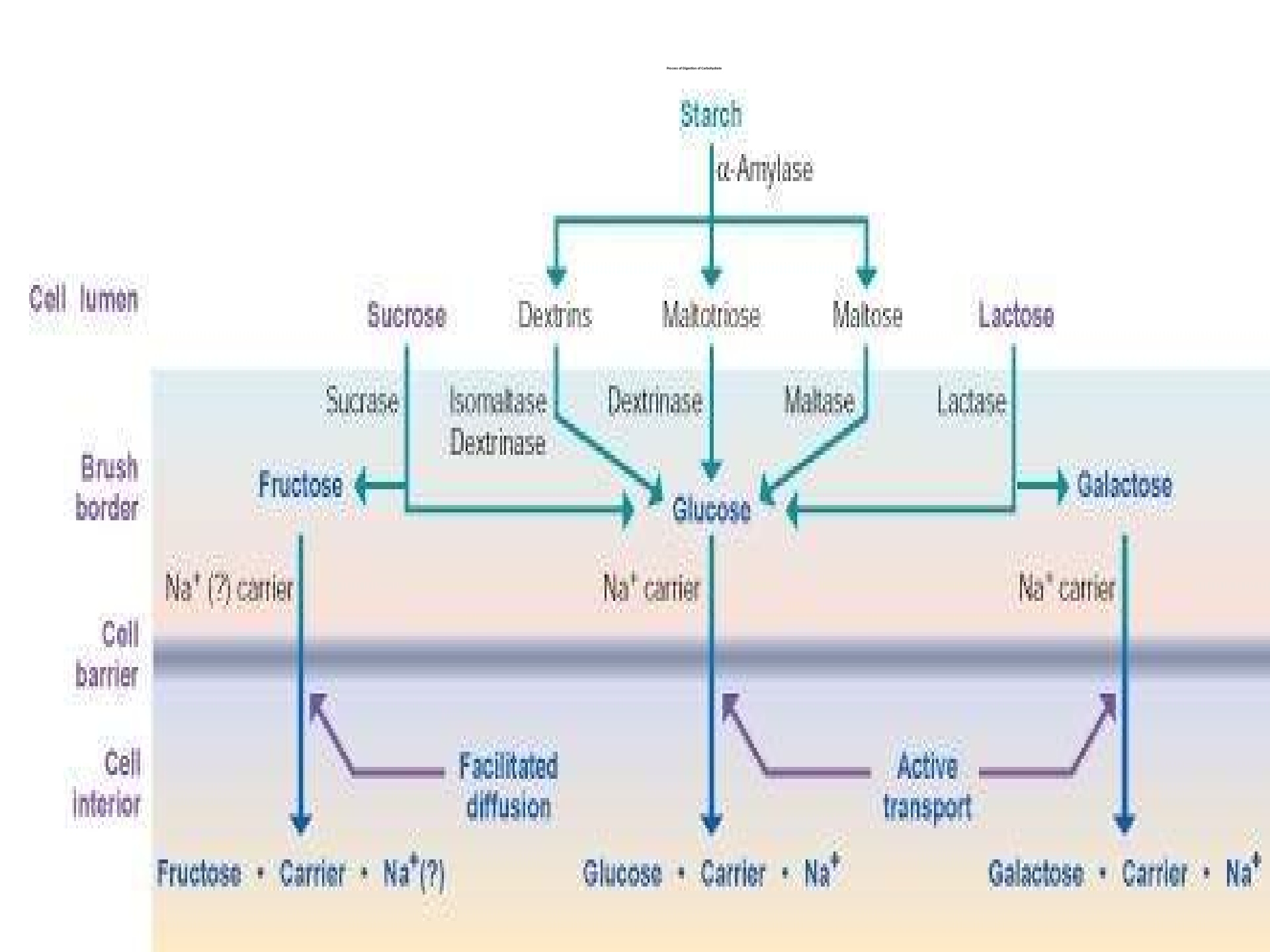
Monosaccharides, the end products of carbohydrate digestion, enter the capillaries of the intestinal villi.




Brush border membrane
Basolateral membrane
Intestinal lumen
Enterocyte
Portal blood
Glucose
Glucose
SGLT1
2 Na+
2 Na+
Glucose, galactose, or fructose
Glucose, galactose, or fructose
GLUT2
Fructose
Fructose
GLUT5
ATP
Galactose
Galactose
2 K+
2 K+
SGLT1
3 Na+
3 Na+
2 Na+
2 Na+
ADP + Pi

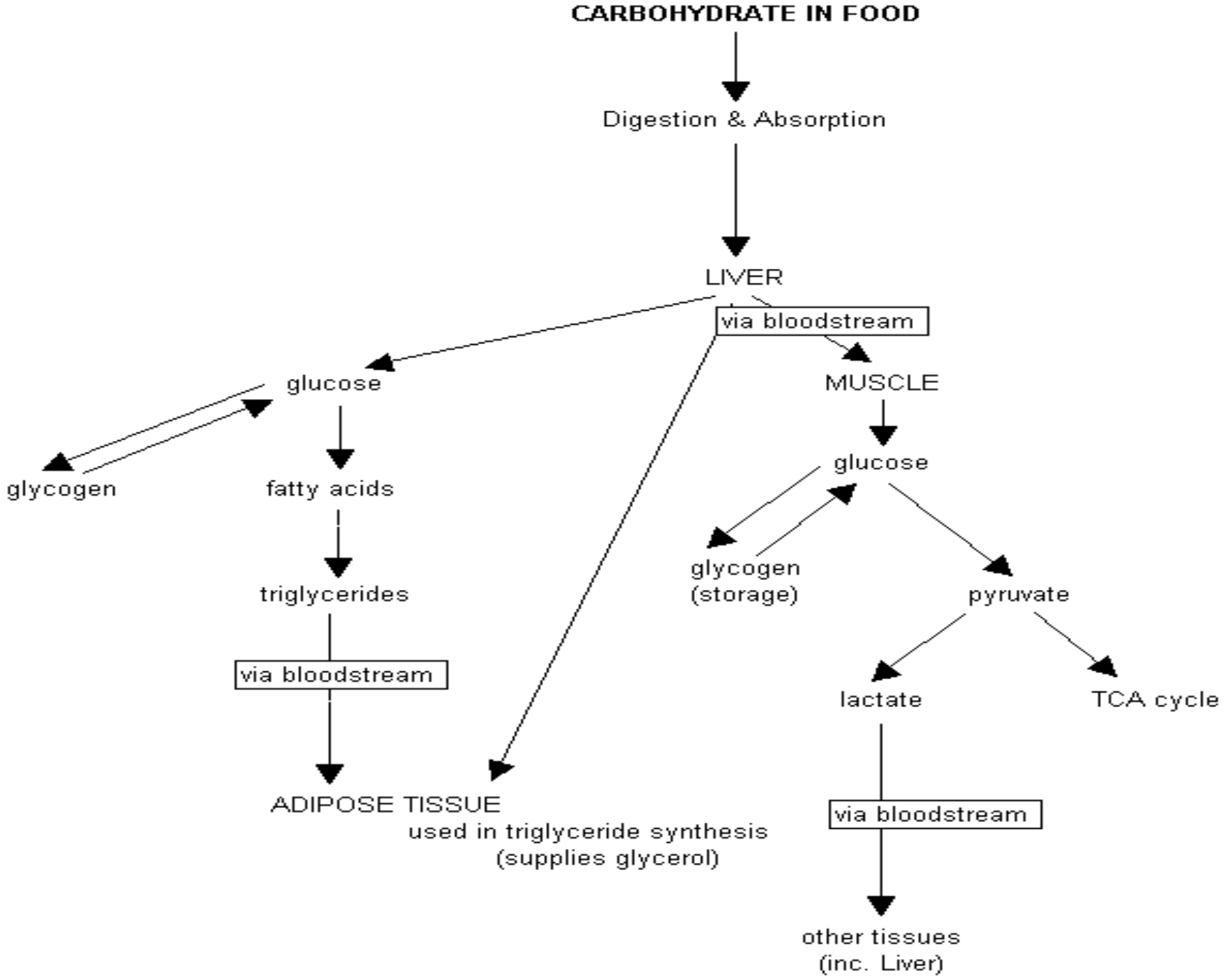


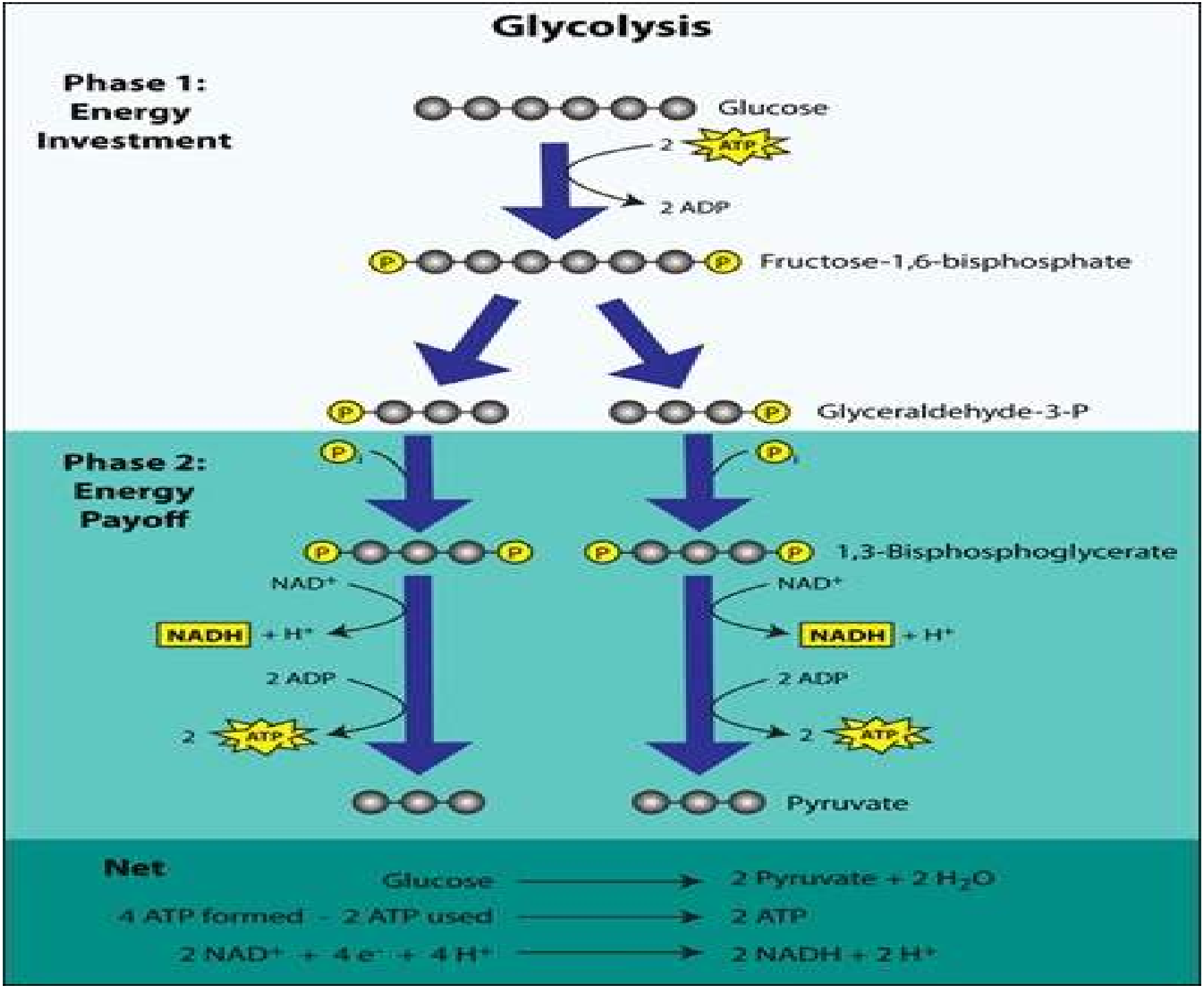
Summary of carbohydrate digestion in the human body.


Process of Digestion of Carbohydrate
Starch
α-Amylase
Cell lumen
Sucrose
Dextrins
Maltotriose
Maltose
Lactose
Sucrase
Isomaltase
Dextrinase
Dextrinase
Maltase
Lactase
Brush border
Fructose
Glucose
Galactose
Na+ (?) carrier
Na+ carrier
Na+ carrier
Cell barrier
Cell interior
Facilitated diffusion
Active transport
Fructose • Carrier • Na+(?)
Glucose • Carrier • Na+
Galactose • Carrier • Na+

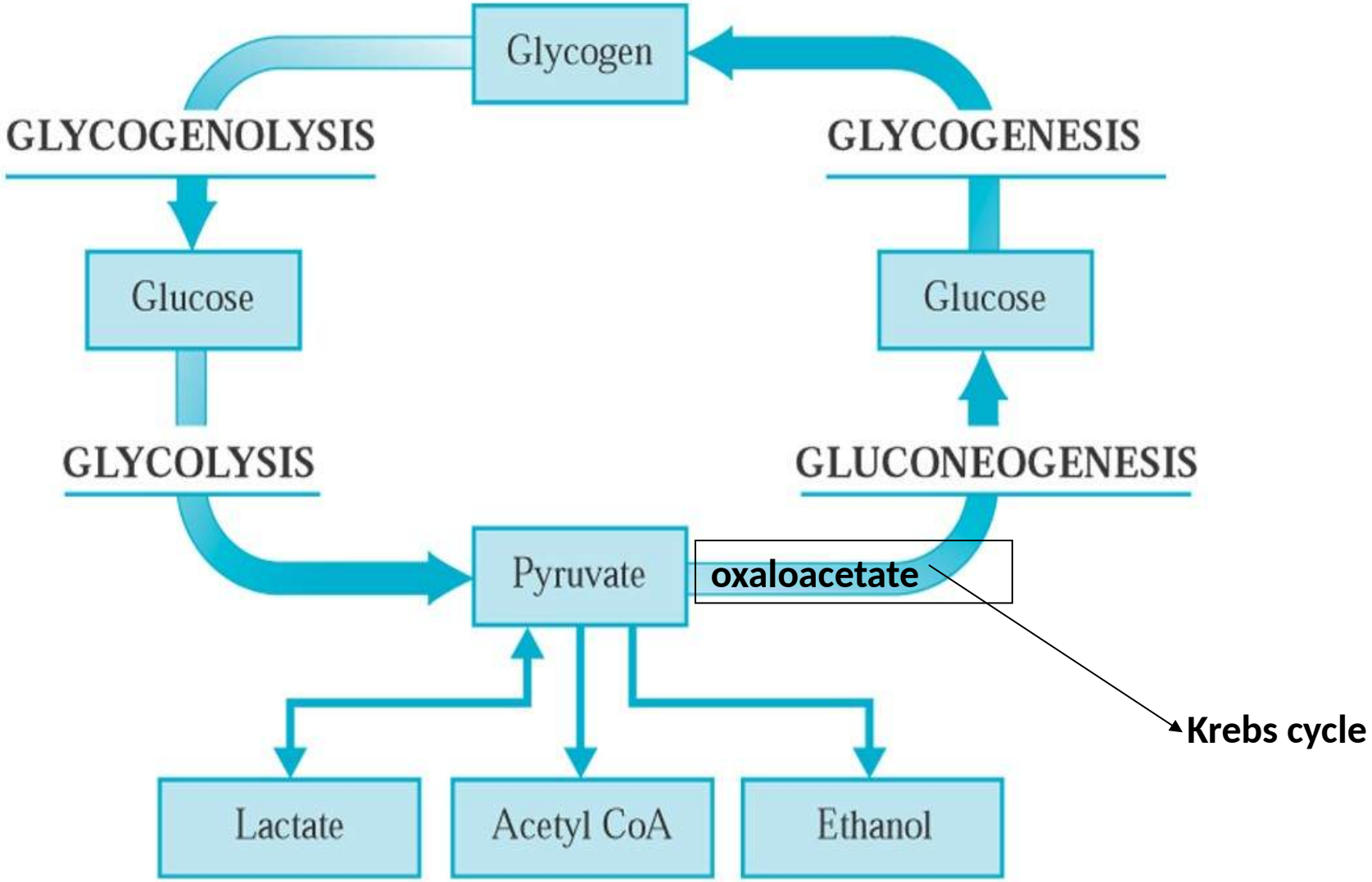
Metabolism of carbohydrates

- Chemical reactions that occur in the body:
  - Building and maintaining body tissues
  - Regulating body functions
  - Supplying energy
- For metabolism to occur the body needs:
  - Water
  - Energy
  - Oxygen
  - Nutrients

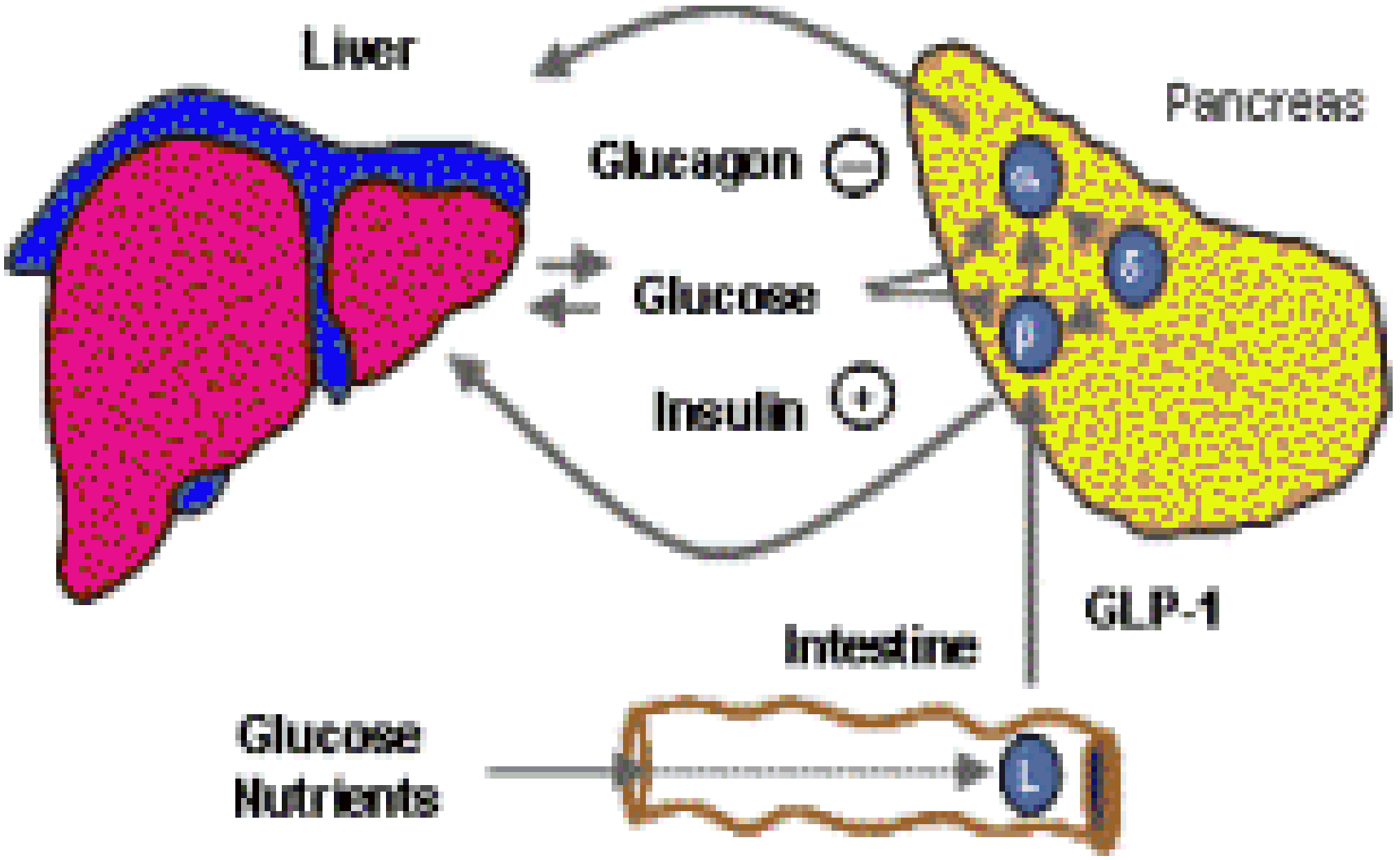

CARBOHYDRATE IN FOOD
Digestion & Absorption
LIVER
via bloodstream
glucose
MUSCLE
glycogen
fatty acids
glucose
glycogen
(storage)
pyruvate
triglycerides
via bloodstream
lactate
TCA cycle
ADIPOSE TISSUE
used in triglyceride synthesis
(supplies glycerol)
via bloodstream
other tissues
(inc. Liver)


Glycolysis
Phase 1: Energy Investment
Glucose
2
ATP
2 ADP
Fructose-1,6-bisphosphate
Glyceraldehyde-3-P
Phase 2: Energy Payoff
1,3-Bisphosphoglycerate
NAD+
NADH + H+
2 ADP
2
ATP
Pyruvate
Net
Glucose → 2 Pyruvate + 2 H2O
4 ATP formed − 2 ATP used → 2 ATP
2 NAD+ + 4 e− + 4 H+ → 2 NADH + 2 H+

The relationships among four common metabolic pathways that involve glucose.

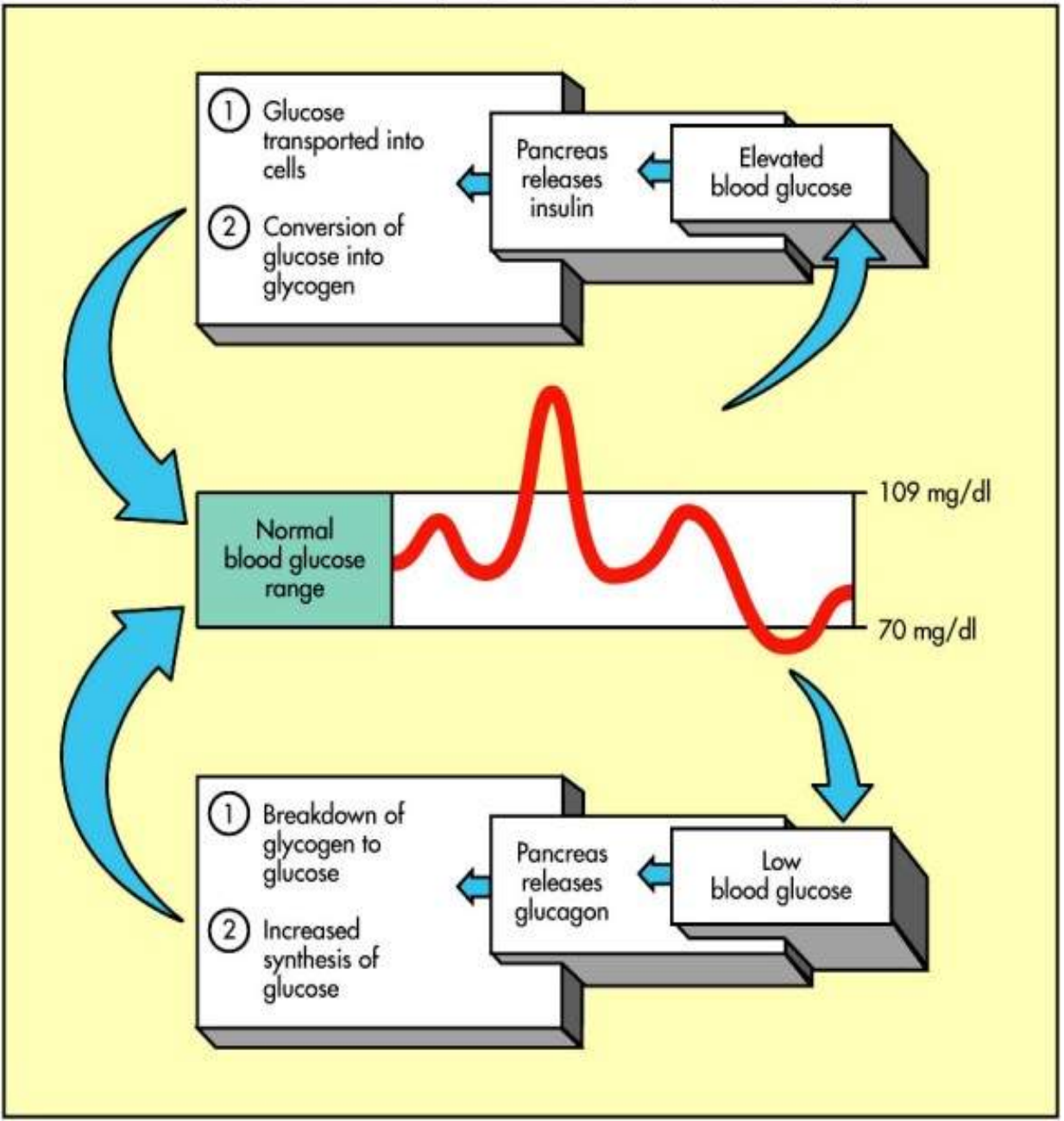


Glucose Metabolism



How does the body respond to glucose?

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Hyperglycemia

Homeostasis

Hypoglycemia

Other Hormonal Influence

- Epinephrine/norepinephrine
  - “fight or flight” response
  - breakdown glycogen
  - raises blood glucose
- Cortisol and growth hormone
  - increase gluconeogenesis
  - raise blood glucose

A Preview of Carbohydrate Metabolism

- The body stores glucose as glycogen in liver and muscle cells.
- The body uses glucose for energy if glycogen stores are available.
- If glycogen stores are depleted, the body makes glucose from protein.
  - Gluconeogenesis is the conversion of protein to glucose.
  - Protein-sparing action is having adequate carbohydrate in the diet to prevent the breakdown of protein for energy.

A Preview of Carbohydrate Metabolism

– Making ketone bodies from fat fragments
  - The accumulation of ketone bodies in the blood is called ketosis.
  - Ketosis upsets the acid-base balance in the body.
– The body can use glucose to make body fat when carbohydrates are consumed excessively.

Summary of Enzymatic Digestion and Absorption

Secretion	Enzyme	Substrate	Action	Final Product
Saliva	Ptyalin	Starch	Hydrolysis to form dextrins	
Gastric juice	Pepsin	Protein	Hydrolysis of peptide bonds	
	Gastric lipase	Fats	Hydrolysis into free fatty acids	

Summary of Enzymatic Digestion and Absorption —cont'd

Secretion	Enzyme	Substrate	Action	Final Product
Pancreatic exocrine secretion	Lipase	Fat	Hydrolysis to mono-glycerides	Fatty acids
	Cholesterol esterase	Cholesterol	Hydrolysis to esters of cholesterol and fatty acids	Choles-terol
	alpha-Amylase	Starch, dextrins	Hydrolysis	Dextrin, maltose

Summary of Enzymatic Digestion and Absorption —cont'd

Secretion	Enzyme	Substrate	Action	Final Product
Pancreatic exocrine secretion	Trypsin	Protein	Hydrolysis	Polypeptides
	Chymotrypsin	Protein	Hydrolysis	Polypeptides
	Carboxy-peptidase	Polypep-tides	Hydrolysis	Amino acids
	Ribonuclease	Ribonu-cleic acids	Hydrolysis	Mono-nucleotides
	Elastase	Fibrous protein	Hydrolysis	Amino acids

Summary of Enzymatic Digestion and Absorption —cont'd

Secretion	Enzyme	Substrate	Action	Final Product
Brush border enzymes	Carboxy-peptidase; aminopep-tidase; dipeptidase	Polypep-tides	Hydrolysis	Amino acids
	Entero-kinase	Trypsino-gen	Activates to trypsin	Polypep-tidases and peptides
	Sucrase	Sucrose	Hydrolysi	Glucose,

Enzymatic Digestion and Absorption —cont'd

Secretion	Enzyme	Substrate	Action	Final Product
Brush border enzymes	Isomaltase	Dextrin	Hydrolysis	Glucose
	Maltase	Maltose	Hydrolysis	Glucose
	Lactase	Lactose	Hydrolysis	Glucose, galactose
	Nucleotidases	Nucleic acid	Hydrolysis	Nucleotides
	Nucleosidases and	Nucleosidases	Hydrolysis	Purine and pyrimidine

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